

Lujain Al-Momani

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[linkedin.com/in/lujain-almomani](https://www.linkedin.com/in/lujain-almomani) | [Laj-22.github.io/lujainMomani-portfolio/](https://github.com/Laj-22) | Legal Status:Golden Visa Holder

Education

University of Wollongong in Dubai, Bachelor of Engineering in Telecommunication & IoT Dubai, UAE

Experience

Research Intern, University of Wollongong (UOWD) – Dubai, UAE Sept 2024 – Present

- Developed an ML-based Intrusion Detection System (IDS) for VANETs with 97% accuracy, leading to published research on vehicular network security.
- Implemented system optimization techniques for real-time threat detection in resource-constrained environments.

Clubs Director - Student Government, Rochester Institute of Technology (RIT) – Dubai, UAE Apr 2023 – Sept 2023

- Led university clubs and organized campus-wide events, earning nominations for Best Leadership and Best Club.
- Facilitated collaboration between students, faculty, and staff to strengthen community engagement.

Business Intelligence Intern, Jordan Payments & Clearance Company (JOPACC) – Amman, Jordan July 2023 – Aug 2023

- Developed a key presentation and quarterly report to support strategic business decisions.
- Gained insights into ETL processes while utilizing Power BI for data visualization and reporting improvements.

Publications

- Almomani, L., et al. **A Hybrid FeedForward Neural Network-Genetic Algorithm Approach for Beam Selection Optimization in 5G Massive MIMO Systems**. ICCE 2025.
- Almomani, L., et al. **Beyond Manual Labelling: Enhancing Antenna Classification and Performance with Fuzzy Clustering and Machine Learning**. FICTA 2025.
- Abbasi, M., Almomani, L., Alkhatib, O. **Intrusion Detection in VANETs: A Comparative Study of Machine Learning and Deep Learning Models**. ISDIA 2025.
- Almomani, L., et al. **A Comprehensive Review of Security Mechanisms in VANETs and IoV**. (Under Review)
- Abbasi, M., Almomani, L., Alkhatib, O. **Real-Time Intrusion Detection in VANETs Using Lightweight Machine Learning and MQTT**. (Under Review)

Projects

Hardware-Accelerated Intrusion Detection System for VANETs — Prototyped a real-time, hardware-based IDS on a Raspberry Pi 5, achieving 99.72% detection accuracy and validating optimal communication protocols.

Autonomous Waste Collection Robot (ERC 2025) — Engineered an autonomous UGV by integrating ROS navigation, a custom forklift mechanism, and a fine-tuned MobileNetV2 model for real-time object detection.

Smart IoT Parking Management System (SmartSpot) — Developed an end-to-end smart parking solution with an ESP32, ultrasonic sensors, and a robust Number Plate Recognition pipeline (92% accuracy), streaming real-time data to a cloud dashboard.

Multi-Site Enterprise Network Design — Designed and configured a secure, scalable network connecting two offices using a hierarchical star topology, a site-to-site IPsec VPN, and a multi-zone security architecture.

Skills & Certifications

Networking, Cybersecurity & Management: Network Architecture & Security, Intrusion Detection Systems (IDS), Fault-Tolerant System Design, Network Protocols, Agile Project Management, Process Optimization, Technical Research, Strategic Planning, Risk Assessment, Team Leadership.

Languages & Technologies: Python, Java, C++, MATLAB, ROS, Git, VMware, Kali Linux, Cisco IOS, KiCad, Machine Learning.

Certifications: Agile Software Development (Nano Degree, 2023), McKinsey Forward Program (2024), PWC Elevate Me (2025).

Languages: Arabic, English || **Affiliations:** IEEE, Engineers Australia